

Hardware Projects [Project Source](#)

Design, Build, Analyze, and Learn

1. HYGIENE PRODUCT DISPENSER

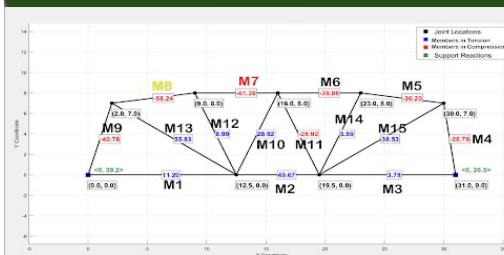
Gesture-Controlled Dispenser



- ✓ Three gesture controls and proximity control for pads, tampons, and toilet paper
- ✓ Response time: < 3 seconds
Low power
- ✓ Three stepper motors, LEDs, Arduino, and APDS 9960 gesture sensor

2. TRUSS DESIGN PROJECT

Maximizing Strength and Efficiency



- ✓ Max theoretical load: 65.698
- ✓ Critical member: #7 (7in) | Buckling strength: 61.26 oz
- ✓ Design created, tested, and validated
- ✓ Optimized geometry, even distribution of forces, and rounded top for improved performances

3. DIGITAL THERMOMETER

Real-Time Temperature Monitoring



- ✓ Temperature sensor with live readings
- ✓ LCD display for real-time monitoring
- ✓ Arduino-controlled embedded system
- ✓ Audible alarm when temperature falls outside the 68-75 degrees fahrenheit range